Do **not** write solutions on this page.

11. [Maximum mark: 18]

(a) Use integration by parts to find $\int \arccos x \, \mathrm{d}x$. [4]

The probability density function of a random variable X, is given by

$$f(x) = \begin{cases} 3x\arccos\left(x^2\right), & 0 \le x \le k; \\ 0, & \text{otherwise} \end{cases}.$$

(b) (i) Show that $k^2 \arccos(k^2) - \sqrt{1-k^4} + \frac{1}{3} = 0$.

(ii) Hence, find the value of k. Give your answer correct to six significant figures. [6]

(c) Find

(i) $\mathrm{E}(X)$;

(ii) $\mathrm{Var}(X)$. [5]

(d) Given that $\mu = \mathrm{E}(X)$ and $\sigma^2 = \mathrm{Var}(X)$, find $\mathrm{P}(\mu - \sigma < X < \mu + \sigma)$. [3]